

KITACO 4 x 4 MS Centrifugal Pump coupled to HATZ 1B40U-X Diesel Engine

Test Report No.	:	2013 - 03
Date of Test	:	February 3, 2013
Test Valid Until	:	February 3, 2018
Brand	:	KITACO
Model	:	4 x 4 MS
Serial No.	:	No Data
Type and Size	:	Single-end suction, 100mm x 100mm, Centrifugal pump
Manufacturer	:	Made in China
Test Requested by	:	M & W Distributing Co., Inc. 14 Panay Ave., Quezon City
Machine Classification	:	Commercial

SUMMARY

The KITACO 4 x 4 MS Centrifugal Pump coupled to HATZ 1B40U-X diesel engine was tested for performance and suction condition. It was tested at the recommended full throttle speed setting of the engine.

At zero discharge condition, the Maximum Total Head (H) was 23.5 m. The shaft speeds of the pump and engine at this point were 2134 and 2732 rpm, respectively, with Engine Fuel Consumption Rate of 1.04 L/h. The Maximum Discharge Capacity of 28.8 L/s occurred at 8.71 m Total Head with Fuel Consumption rate of 1.74 L/h. At this point, the shaft speeds of the pump and engine were 1583 rpm and 2346 rpm respectively.

The Maximum Fuel Consumption Rate of 1.74 L/h occurred at the point of Maximum Discharge Capacity.

The suction condition of the pump was also tested. With zero discharge, the Maximum Total Suction Lift was 7.93 m. The Net Positive Suction Head Available at this point was 1.98 m.

PURPOSE AND SCOPE OF TEST

The purpose of the test was to establish the pumpset performance which included the Fuel Consumption Rate and the Total Head at varying discharge capacities and suction conditions. **Figure 1** shows the pump mounted on the laboratory test set-up.

The test was conducted on February 3, 2013 at the AMTEC Test Laboratory, UPLB, College, Laguna.

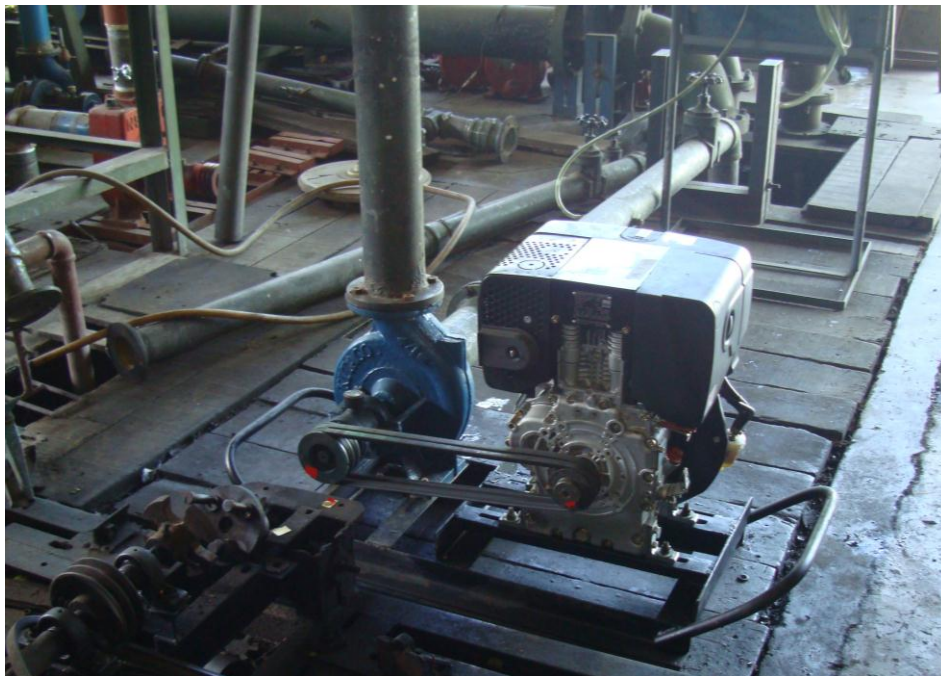


Figure 1. KITACO 4 x 4 MS Centrifugal Pump coupled to HATZ 1B40U-X Diesel engine on the test set-up

METHOD OF TEST

The pump was tested following the “Philippine Agricultural Engineering Standards, PAES 115:2000, Agricultural Machinery – Centrifugal, Mixed-Flow, Axial-Flow Water Pumps – Methods of Test”.

DESCRIPTION OF THE PUMP

The KITACO 4 x 4 MS shown in **Figure 2**, was a single-end suction, 100 x 100 mm, self-priming centrifugal pump coupled to a 7.7 kW (10.3 hp) HATZ 1B40U-X air-cooled diesel engine. The pump and engine were mounted on a steel base. The power from the engine was transmitted to the pump shaft through a double groove B-section pulley and V-belts. The actual diameter of the engine and pump pulleys were 76.2 mm (3.0 inches) 101 mm (4.0 inches). The suction inlet and discharge outlet were threaded flanges and designed for a 100 mm diameter pipe. The impeller was closed type made of brass material connected to the pump shaft through a thread. It was housed in a cast iron volute casing provided with priming inlet. The discharge outlet was located on top of the pump facing upward, while the suction inlet was placed in-line with the centerline of the pump. The stuffing box used a mechanical seal. The pump shaft rotated on ball bearings kept in place by a bearing housing integral to the casing.

The primemover, a HATZ 1B40U-X Diesel engine, was a 7.7 kW, four-stroke, air-cooled, single vertical cylinder diesel engine. Fuel feed system was of gravity type with the fuel tank located beside the engine. It utilized a dry type air cleaner using pleated paper. Lubrication system was force-feed type. Cooling was done by air forced by a fan or flywheel encased in a metal shroud. It was started by a rope-recoil starting system. A muffler was used in the exhaust system.

Table 1 shows the detailed specifications of the pumpset.

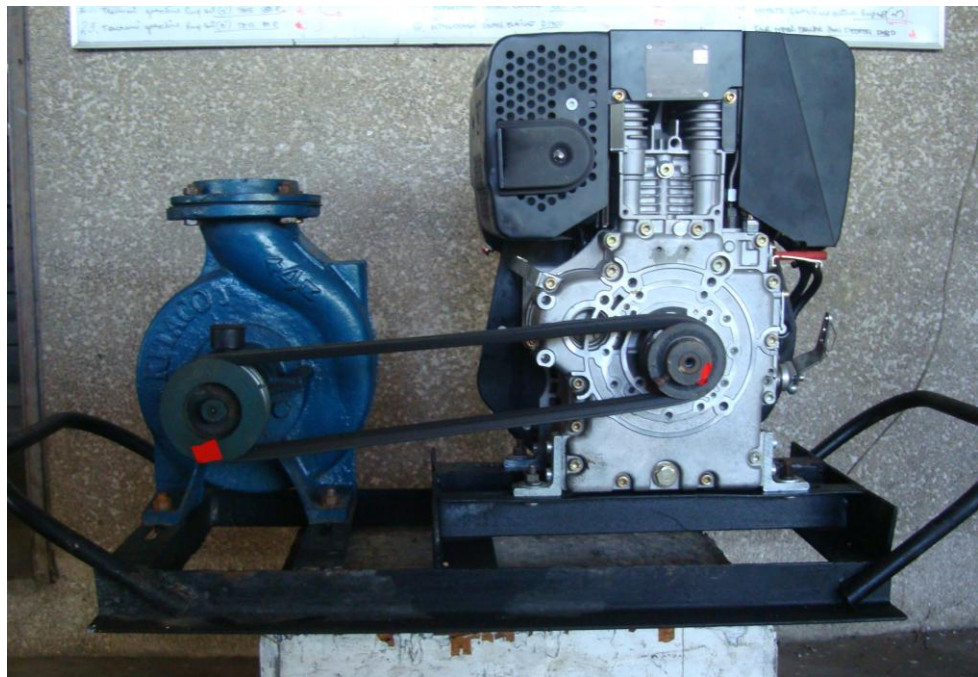


Figure 2. KITACO 4 x 4 MS Centrifugal Pump coupled to a HATZ 1B40U-X Diesel Engine

PUMPSET SPECIFICATIONS

Table 1. Detailed specifications of the KITACO 4 x 4 MS Centrifugal Pump coupled to a HATZ 1B40U-X Diesel Engine.

1.	Brand	:	KITACO
2.	Model	:	4 x 4 MS
3.	Serial No.	:	No Data
4.	Make	:	Made in China
5.	Type	:	Single-end suction, Centrifugal pump
6.	Size (mm)	:	100 x 100
7.	Overall Dimension and Weight (Including mounting frame and engine)		
7.1	Length (mm)	:	1110
7.2	Width (mm)	:	460
7.3	Height (mm)	:	590
7.4	Weight (kg)	:	160.0
8.	Impeller		
8.1	Type	:	Closed
8.2	Diameter (mm)	:	201
8.3	Width (mm)	:	16.9
8.4	No. of vanes	:	9
8.5	Material	:	Brass
9.	Suction Inlet		
9.1	Type	:	Threaded flange
9.2	Diameter (mm)	:	100
9.3	Material	:	Cast Iron

Table 1. (Cont'n)

10.	Discharge Outlet		
10.1	Type	:	Threaded flange
10.2	Diameter (mm)	:	100
10.3	Material	:	Cast Iron
11.	Casing		
11.1	Type	:	Volute
11.2	Material	:	Cast Iron
12.	Stuffing box	:	Mechanical Seal
13.	Rotation	:	Clockwise when facing the suction inlet
14.	Rated speed (rpm)	:	No data available
15.	Rated Maximum Discharge Capacity (L/s)	:	No data available
16.	Rated Maximum Total Head (m)	:	No data available
17.	Primemover		
17.1	Brand	:	HATZ
17.2	Model	:	1B40U-X
17.3	Serial No.	:	1101612 041225
17.4	Make	:	Made in Germany
17.5	Type	:	Four stroke, air-cooled single vertical cylinder cylinder diesel engine
17.6	Rated Output Power (kW)		
	Maximum	:	7.7 @ 3600 rpm
	Continuous	:	No Data

Table 1. (Cont'n)

17.8	Bore x Stroke (mm)	:	88 x 76
17.9	Displacement Volume (cc)	:	462
17.10	Cooling System	:	Air-cooled
17.11	Lubricating System	:	Force-feed type
17.12	Starting System	:	Rope-recoil
17.13	Air Cleaner	:	Dry (Pleated paper element)s
17.14	Exhaust System	:	Muffler

PERFORMANCE TEST

Results of performance test are shown in **Table 2** and plotted in **Figure 3**.

Pump Tested : KITACO 4 x 4 MS coupled to
HATZ 1B40U-X Diesel Engine

Date of Test : February 3, 2013

Test Condition

Ambient Temperature (°C) : 28.1
Atmospheric Pressure (mb) : 1013
Pumping Water Temperature (°C) : 29.0

Table 2. Results of Performance Test

ENGINE SHAFT SPEED (rpm)	PUMP SHAFT SPEED (rpm)	FUEL CONSUMP- TION (L/h)	TOTAL HEAD (m)	DISCHARGE CAPACITY (L/s)	WATER POWER (kW)	INPUT POWER (kW)	SYSTEM EFFICIENCY (%)
2732	2134	1.04	23.5	0	0	10.8	0
2722	2079	1.20	22.7	2.44	0.54	12.5	4.35
2700	2012	1.30	21.6	6.08	1.29	13.5	9.52
2690	1951	1.37	20.0	8.69	1.71	14.3	12.0
2677	1879	1.47	17.6	13.4	2.30	15.3	15.1
2657	1820	1.59	15.2	18.6	2.76	16.5	16.7
2631	1783	1.66	12.7	23.1	2.88	17.3	16.6
2599	1672	1.69	10.1	27.5	2.73	17.5	15.6
2510	1621	1.72	8.90	28.5	2.49	17.9	13.9
2346	1583	1.74	8.71	28.8	2.46	18.1	13.6

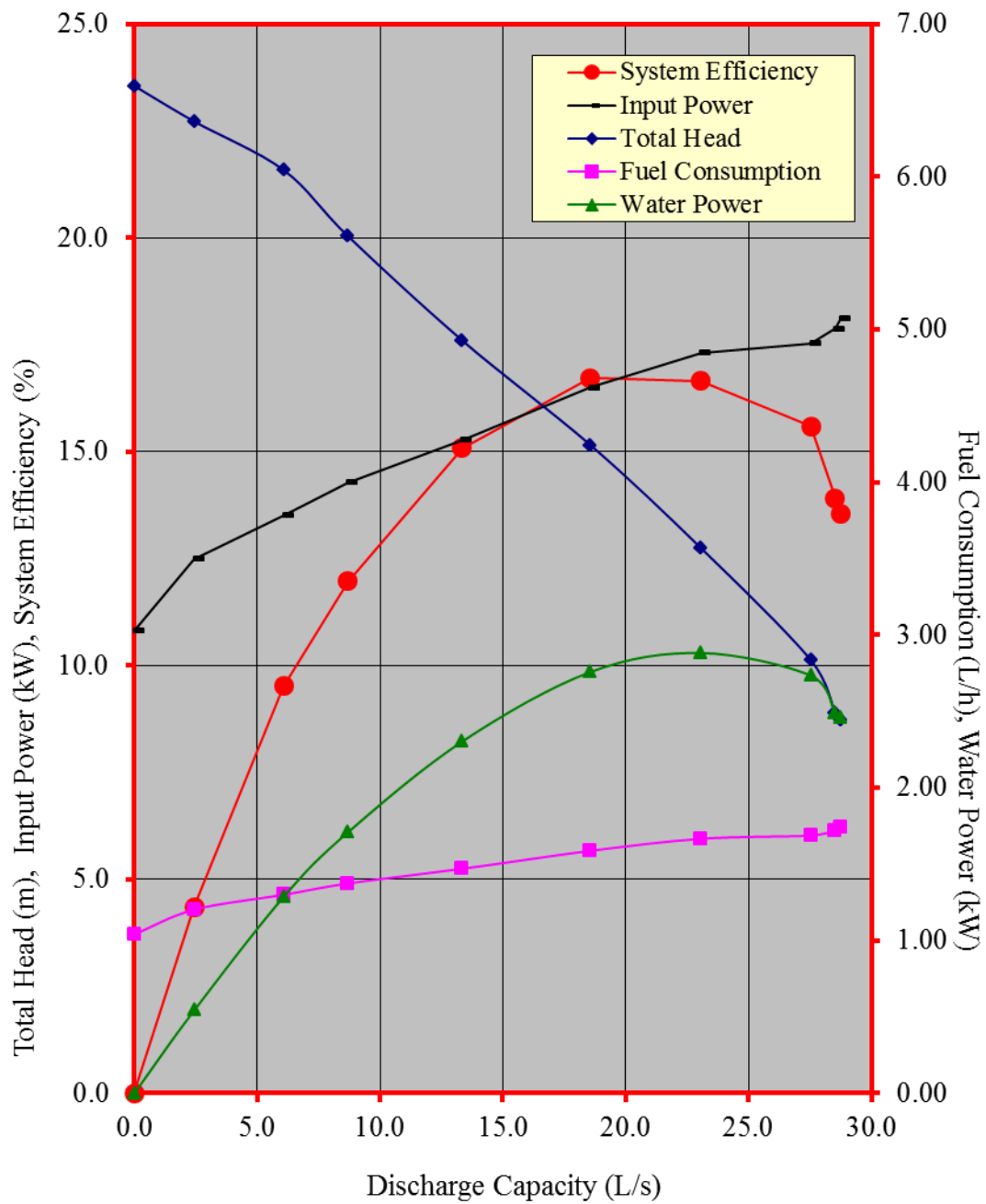


Figure 3. Performance characteristic curves of KITACO 4 x 4 MS Centrifugal Pumpset coupled with HATZ 1B40U-X Diesel Engine

CAVITATION TEST

Results of Cavitation Test are shown in **Table 3**.

Pump Tested : KITACO 4 x 4 MS Pump coupled to
HATZ 1B40U-X diesel engine

Date of Test : February 3, 2013

Test Condition

Ambient Temperature (°C) : 28.1
Atmospheric Pressure (mb) : 1013
Pumping Water Temperature (°C) : 29

Table 3. Results of Cavitation Test

ENGINE SHAFT SPEED (rpm)	PUMP SHAFT SPEED (rpm)	TOTAL SUCTION LIFT (m)	NET POSITIVE SUCTION HEAD AVAILABLE (m)	TOTAL HEAD (m)	DISCHARGE CAPACITY (L/s)
2698	1924	0.01	9.87	10.6	28.9
2648	1811	0.76	9.13	10.4	27.5
2647	1813	1.37	8.51	10.3	28.0
2646	1819	3.22	6.67	9.46	24.2
2640	1824	4.75	5.14	8.95	22.8
2665	1681	6.76	3.13	6.90	15.9
2740	2091	7.93	1.98	4.86	0

DATA COMPUTATION

ANNEX 1. Formula used during calculation and testing

1. Total Head (m), H

$$TDH = H_d - H_s$$

$$\text{Where: } H_d = \frac{P_2}{\gamma} + \frac{V_2^2}{2g} + Z_2$$

$$H_s = \frac{P_1}{\gamma} + \frac{V_1^2}{2g} + Z_1$$

H_d = Total Discharge Head (m)

H_s = Total Suction Lift/Head (m)

$\frac{P_2}{\gamma}$ = Pressure gage reading on the discharge pipe at
gage connection converted to meter

$\frac{P_1}{\gamma}$ = Vacuum gage reading on the suction pipe at the
point of gage connection converted to meter

$\frac{V_2^2}{2g}$ = Velocity head on the discharge pipe converted
to meter. Computed by determining the velocity
of liquid on the pipe by formula, $V = Q/A$ where
 V , velocity of liquid inside the pipe in meter per
per second, Q , Discharge rate of the pump in
liters per second, and A , Cross-sectional area
of the pipe in square meter.

$\frac{V_1^2}{2g}$ = Velocity head on the suction pipe converted to
meter

Annex (Cont'n)

Z_2 = The distance from the center of the pressure gage face

to the centerline of the pump in meters

Z_1 = The distance from the center of the vacuum gage face
to the centerline of the pump in meters

2. Discharge Capacity, Q , in liters per second (L/s) is determined through gravimetric method using a platform scale and a stopwatch.
3. Water Power, WP , is the output power of the pump in kilowatt (kW) and computed using:

$$WP = \frac{H \times Q}{102}$$

4. Input Power , IP , is the power on the input shaft of the pump. This is the power in kilowatt required to drive the pump and is determine by the formula:

$$IP = \frac{T \times N}{974}$$

Where:

T = Torque in kilogram-meter, on the input shaft of the pump and determined by a dynamometer

N = Input shaft angular speed in RPM

5. Pump Efficiency, η ,

$$\eta = \frac{WP}{IP}$$

6. Net Positive Suction Head Available, NPSHA. This is the net force (converted to meter) needed to push the water inside the pipe and computed by the formula:

$$NPSHA = P_{atm} - P_{vp} - H_s$$

Where:

P_{atm} = Atmospheric pressure during the test converted to meters

P_{vp} = Vapor pressure of the test liquid at certain temperature
converted to meters

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N.B.

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